

Battery Design P55 - Shay, Michael, Mark

Problem 3.3

a) Pure lead

Electrode held @ $V = -0.44 \text{ V}$ vs SHE, $T = 25^\circ\text{C}$

$$F/RT = 38.92 \text{ V}^{-1}$$

$$\eta_i = V - U_i^\theta$$

PbSO_4 rxn desired

$$\eta_{\text{Pb}} = -0.44 - (-0.356) = 0.084 \text{ V}$$

Butler-V w/ $\alpha = 0.5$:

$$i_{\text{Pb}} = 100 \left[e^{(0.5)(38.92)(-0.084)} - e^{-(0.5)(38.92)(-0.084)} \right] \rightarrow$$

$$= 100 \left[e^{-1.635} - e^{+1.635} \right] = 100 (0.195 - 5.13)$$

$$i_{\text{Pb}} = -493 \text{ A}\cdot\text{m}^{-2} \text{ (Cathodic - charging)}$$

H_2 side rxn

$$\eta_{\text{H}_2} = -0.44 - 0 = -0.44 \text{ V}$$

$$i_{\text{H}_2} = 6.6 \times 10^{-10} \left[e^{-8.56} - e^{+8.56} \right] \approx 6.6 \times 10^{-10} (-5228)$$

$$i_{\text{H}_2} = -3.45 \times 10^{-6} \text{ A}\cdot\text{m}^{-2} \text{ (negligible in pure Pb)}$$

b) w/ Sb impurity ($i_{0, \text{H}_2} = 3.7 \times 10^{-4} \times (-5228) = -1.93 \text{ A}\cdot\text{m}^{-2}$)

	Pure Pb	w/ Sb
i_{PbSO_4}	$-493 \text{ A}\cdot\text{m}^{-2}$	$-493 \text{ A}\cdot\text{m}^{-2}$
i_{H_2}	$-3.5 \times 10^{-6} \text{ A}\cdot\text{m}^{-2}$	$-1.93 \text{ A}\cdot\text{m}^{-2}$
H_2	0%	0.4%

Trace amounts of Sb contamination raises H_2 gases from negligible to traceable. When charging -0.4% of current goes to H_2O electrolysis, causing electrolyte H_2O loss & requiring maintenance.

Problem 3.4

$$i_0 = A \exp\left(-\frac{E_a}{RT}\right) \rightarrow \ln i_0 = \ln A - \frac{E_a}{R} \cdot \frac{1}{T}$$

T/°C	T(K)	1/T × 10 ³ (K ⁻¹)	ln(i ₀)
15	288	3.472	-14.18
25	298	3.356	-13.29
35	308	3.247	-11.79
45	318	3.145	-11.21

$$\ln(i_0) \text{ vs } \frac{1}{T} \text{ slope} = 9548 \text{ K}$$

slope $E_a = 95 \times R = 9548 \times 8.314 = 79,382 \rightarrow 79.4 \text{ kJ mol}^{-1}$

$$A = 1.66 \times 10^8 \text{ A m}^{-2}$$

intercept $i_0(T) = 1.48 \times 10^8 \exp\left(\frac{9548}{T}\right) \text{ A m}^{-2}$

$$T \text{ for } i = 5 \text{ A m}^{-2} \text{ (O}_2 \text{ reduction, } |n| = 0.7 \text{ V)}$$

Tafel approx

$$|i| = i_0(T) \exp\left(\frac{aF|n|}{RT}\right) = 5 \text{ A m}^{-2}$$

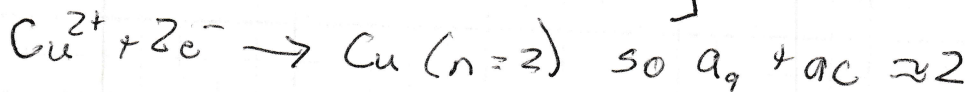
$$\begin{aligned} 5 &= 1.66 \times 10^8 \exp\left(-\frac{9548}{T} + \frac{0.5 \times 96485 \times 0.7}{8.314 T}\right) \rightarrow \\ &= 1.66 \times 10^8 \exp\left(\frac{-9548 + 4061}{T}\right) \exp\left(-\frac{5487}{T}\right) = \frac{5}{1.66 \times 10^8} \rightarrow \\ &= 3.01 \times 10^{-8} \end{aligned}$$

$$T = \frac{5487}{\ln\left(\frac{1}{3.01 \times 10^{-8}}\right)} = \frac{5487}{17.32} = 317 \text{ K} = 44^\circ \text{C}$$

Problem 3.10

Data spans both Tafel regions & the linear region near eq, allowing extraction of $i_0, a_a, + a_c$

$$i = i_0 \left[e^{(a_a F_n / RT)} - e^{(-a_c F_n / RT)} \right]$$



Linear area ~~area~~ $A + n_s = 0.00101 \text{ V}$, $i = 2.3 \text{ A m}^{-2}$

$$i \approx i_0 \frac{(a_a + a_c) F_n}{RT} = i_0 \frac{n F_n}{RT}$$

$$i_0 = \frac{i RT}{n F_n} = \frac{2.3 \times 8.314 \times 298}{2 \times 96485 \times 0.00101} = \boxed{29.238 \text{ A m}^{-2}}$$

Anode Taf Region ($n > 0.07 \text{ V}$)

$\eta_o(\text{V})$	$i (\text{A m}^{-2})$	$\ln i$
0.0192	40	3.689
0.0414	75	4.317
0.0515	100	4.605
0.0616	125	4.828
0.0717	150	5.011
0.0818	200	5.298
0.0919	250	5.521
0.1004	300	5.704

Regression sums ($n=8$)

$$\sum x = 0.5195, \sum y = 38.973$$

$$\sum x^2 = 0.038888$$

$$\sum xy = 2.6570$$

Slope

$$\frac{8(2.6570) - (0.5195)(38.973)}{8(0.038888) - (0.5195)^2}$$

$$\frac{21.256 - 20.244}{0.3110 - 0.26988} = 24.5 \text{ V}^{-1}$$

$$a_a = \frac{24.5}{38.92} = \boxed{0.63}$$

Cathodic data ($\ln i$)

$\eta_o(\text{V})$	$i (\text{A m}^{-2})$	$\ln(i)$
0.0192	67	4.205
0.0293	135	4.905
0.0394	250	5.521
0.0495	450	6.109
0.0657	1100	7.003

Intercept

$$\frac{38.973 - (24.5)(0.5195)}{8} = \frac{38.973 - 12.728}{8} = 3.281$$

$$i_0 = e^{3.281} = 26.602 \text{ A m}^{-2}$$

@ $\eta_0 = 0.818 \text{ V}$: $\ln i = 24.5(0.0818) + 3.281 = 2.004 + 3.281 = 5.285$ vs. $\ln(200) = 5.298$

Regression ($n=5$)

$$\sum x = 0.2031, \sum y = 27.743, \sum x^2 = 0.009545, \sum xy = 1.2045$$

$$\text{slope} = \frac{5(1.2045) - (0.2031)(27.743)}{5(0.009545) - (0.2031)^2} = \frac{6.3899}{0.006475} = 60.2 \text{ V}^{-1}$$

$$a_c = \frac{60.2}{38.92} = 1.55$$

Intercept $\frac{27.743 - (60.2)(0.2031)}{5} = \frac{15.516}{5} = 3.103$

$$i_0 = e^{3.103} = 22.265$$

@ $\eta_0 = 0.0495 \text{ V}$: $\ln i = 60.2(0.0495) + 3.103 = 6.083$ vs. $\ln(450) = 6.109$

Method	$i_0 (\text{A m}^{-2})$	Parameter
Linear Region	29.2	—
Anodic Tafel	26.6	$a_a = 0.63$
Cathodic Tafel	22.3	$a_c = 1.55$
Estimate	~26	—

$$a_a + a_c = 0.63 + 1.55 = 2.18 \quad (n=2)$$

$$i = 26 \left[\exp\left(\frac{0.63 F \eta_a}{RT}\right) - \exp\left(\frac{-1.55 F \eta_c}{RT}\right) \right] \text{ A m}^{-2}$$